UNWRAPPING THE STANDARDS

STUDENT EXPECTATION:	VERTICAL ALIGNMENT:
2A.2A- graph the functions $f(x)=\sqrt{x}$, $f(x)=1/x$, $f(x)=x^3$, $f(x)=\sqrt[3]{x}$, $f(x)=b^x$, $f(x)=\|x\|$, and $f(x)=\log_b(x)$ where b is 2, 10, and e, and, when applicable, analyze the key attributes such as domain, range, intercepts, symmetries, asymptotic behavior, and maximum and minimum given an interval **2A.2B- graph and write the inverse of a function using notation such as $f^{-1}(x)$** **2A.2C- describe and analyze the relationship between a function and its inverse (quadratic and square root, logarithmic and exponential), including the restriction(s) on domain, which will restrict its range** **2A.2D- use the composition of two functions, including the necessary restrictions on the domain, to determine if the functions are inverses of each other** **2A.6A- analyze the effect on the graphs of $f(x) = x^3$ and $f(x) = \sqrt[3]{x}$ when $f(x)$ is replaced by $af(x)$, $f(bx)$, $f(x - c)$, and $f(x) + d$ for specific positive and negative real values of a, b, c, and d** **2A.6B- solve cube root equations that have real roots** **2A.7H- solve equations involving rational exponents**	***Black is from Math beach. Blue is from TEKS Resource***

Highlighted TEKs are considered priority

CONTEXT(S)			
☐ Word Problem ☐ Verbal Description ☐ Chart/Table	☐ Diagram/Image ☐ Number Line ☐ Manipulative/Realia	☐ Equation/Expression ☐ Formula ☐ Geometric Figures	☐ Others:

KEY VOCABULARY		
☐ Composition of functions ☐ Interval notation ☐ Inverse of a function ☐ Cube root equation ☐ Cube root function ☐ Cubic function ☐ Extraneous solution ☐ Horizontal shift	☐ Continuous function ☐ Inequality notation ☐ Range ☐ Relative minimum ☐ x-intercept(s) ☐ Zeros ☐ Horizontal stretch ☐ Horizontal compression ☐ Properties of exponents	☐ Domain ☐ Reflectional symmetry ☐ Relative maximum ☐ y-intercept(s) ☐ Rational exponents ☐ Root ☐ Solution ☐ Vertical compression ☐ Vertical shift ☐ Vertical stretch

SHOW *What will students need to be able to show in their work to demonstrate mastery?*	**KNOW** *What do students need to know to be able to show mastery of this standard?*
Graph functions accurately: Plot $f(x) = x^3$, labeling domain, range, intercepts, and identifying symmetry and end behavior. **Analyze behavior**: Explain the behavior of these graphs, including symmetry, increasing/decreasing intervals, and the lack of asymptotes. **Graph and find inverses**: Given $f(x) = x^3$, students should find and graph its inverse $f^{-1}(x) = \sqrt[3]{x}$, confirming that it reflects over $y = x$. **Inverse notation and relationship**: Demonstrate finding inverses algebraically, using correct notation, and verify the reflection property. **Identify restrictions**: Confirm that cubic functions and cube root functions have all-real-number domains and ranges, and explain situations that might require restrictions. **Composition to verify inverses**: Use function composition to verify if two functions are inverses, ensuring that they satisfy the identity function xxx. **Apply transformations**: Graph and describe transformations of cubic and cube root functions when affected by changes in parameters a, b, c, and d. **Solve and verify solutions**: Solve equations involving cube roots and rational exponents, following proper steps and verifying that solutions are valid within real numbers.	**Graph shapes and behaviors**: Understand the basic graphs and behavior of $f(x) = x^3$, (cubic function) and $f^{-1}(x) = \sqrt[3]{x}$, (cube root function), including their general "S" curve shape for cubic and gradual increase for cube root. **Key attributes**: Recognize that cubic and cube root functions have: • **Domain and Range**: All real numbers. • **Intercepts**: The only intercept at (0,0). • **End Behavior**: Cubic functions extend to positive and negative infinity, cube roots increase gradually. • **Symmetry**: Cubic functions are symmetric about the origin (odd symmetry). **Inverse relationship**: Cube root functions are the inverses of cubic functions, represented as reflections across the line $y = x$. The inverse notation $f^{-1}(x)$ represents cube root functions as inverses of cubic functions. **Domain and Range considerations**: While cubic and cube root functions do not require domain restrictions, understand that other functions may need them to yield real inverses. **Function Composition**: Know that f and g are inverses if $f(g(x)) = g(f(x)) = x$ with any necessary domain restrictions. **Transformation effects**: Understand how transformations using a, b, c, and d in expressions like $af(x), f(bx), f(x - c), f(x) + d,$ affect cubic and cube root functions, including stretching, shrinking, reflecting, and translating. **Solving strategies for equations**: Be familiar with solving cube root equations by isolating the cube root and cubing both sides, and solving equations with rational exponents by using properties of exponents.

POSSIBLE LEARNING TARGETS on level vs advanced….what should be different?
I can graph cubic and cube root functions, identifying key attributes like domain, range, and intercepts. I can describe the symmetry, end behavior, and shapes of cubic and cube root functions. I can find and graph the inverse of a cubic function as a cube root function and explain the relationship between cubic functions and their cube root inverses. I can explain how the domain and range of a function affect its inverse, identifying situations that may require restrictions. I can verify if two functions are inverses by using composition and explain any domain restrictions that apply. I can analyze how changes in parameters a, b, c, and d affect cubic and cube root functions, describing transformations and applying them to real-world contexts. I can solve equations involving cube roots and rational exponents and interpret these solutions in different contexts.